RK3399 System Suspend Configuration Guide

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Preface

Overview

This document guides users on configuring the RK3399 system suspend mode according to product requirements.

Product Version

Chip Models	Kernel Version
RK3399	4.4, 4.19

Intended Audience

This document (this guide) is primarily intended for the following engineers:

Technical Support Engineers

Software Development Engineers

Revision History

Date	Version	Author	Modification Description
2020-07-08	V1.0.0	Xu Shengfei	Initial version

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1. System Suspend

For all SoC platforms equipped with TrustZone, system suspend operations are handled within the TrustZone firmware. Since TrustZone implementations for system suspend vary across different platforms, **suspend configuration options and methods are not correlated or referenceable between platforms; this document applies exclusively to the RK3399 platform**.

The system suspend process typically involves the following operations: powering down power domains, disabling clocks and PLLs, placing DDR into Self-refresh, switching the system bus to a low-speed clock (24M or 32K), commanding the PMIC to enter sleep mode, and configuring wake-up sources. To meet the suspend mode requirements of different products, relevant configurations are currently passed to the TrustZone firmware during the boot phase via Device Tree Source (DTS) nodes.

1.1 Driver Files

```
./drivers/soc/rockchip/rockchip_pm_config.c
./drivers/firmware/rockchip_sip.c
./include/dt-bindings/suspend/rockchip-rk3399.h
```

1.2 DTS Node

```
rockchip_suspend: rockchip-suspend {
    compatible = "rockchip,pm-rk3399";
    status = "okay";
    // Standard configuration
    rockchip,sleep-mode-config = <
        (0
        | RKPM_SLP_ARMPD
        | RKPM_SLP_PERILPPD
        | RKPM_SLP_DDR_RET
        | RKPM_SLP_CENTER_PD
        )
    >;
    // Wakeup source configuration
    rockchip,wakeup-config = <
        (0
        | RKPM_GPIO_WKUP_EN
        )
    >;
    // Power configuration
    rockchip,pwm-regulator-config = <
        (0
        | PWM2_REGULATOR_EN
```

```
        )
    >;
    // Corresponds to APIO power-off
    rockchip,apios-suspend = <
        (0
        | RKPM_APIOxxx
        )
    >;
    // Control the GPIO level during suspend to cut off the corresponding power supply
    rockchip,power-ctrl =
    <&gpioX RK_PXX GPIO_ACTIVE_HIGH>;
};
```

2. DTS Configuration

Currently supported configuration options are defined in:

```
./include/dt-bindings/suspend/rockchip-rk3399.h
```

2.1 Standard Configuration

Configuration items:

```
rockchip,sleep-mode-config = <...>;
```

Configuration source:

```
// The CPU is in WFI state during suspend, used only for debugging
#define RKPM_SLP_WFI                            (1 << 0)
// Suspend: cpu_pd power down
#define RKPM_SLP_ARMPD                          (1 << 1)
// Suspend perilp_pd power down
#define RKPM_SLP_PERILPPD                       (1 << 2)
// During suspend, DDR enters self-refresh and remains in retention state
#define RKPM_SLP_DDR_RET                        (1 << 3)
// Suspend PLL power down
#define RKPM_SLP_PLLPD                          (1 << 4)
// Suspend: OSC disabled, system clock switched to 32K
#define RKPM_SLP_OSC_DIS                        (1 << 5)
// Suspend center_pd power down
#define RKPM_SLP_CENTER_PD                      (1 << 6)
// During suspend, AP_OFF is pulled high to control the PMIC or other discrete power
supplies to enter suspend mode.
#define RKPM_SLP_AP_PWROFF                      (1 << 7)
```

2.2 Power Configuration

Configuration items:

```
rockchip,pwm-regulator-config = <...>;
```

Configuration source:

```
// Use pwm-regulator
#define PWM0_REGULATOR_EN                       (1 << 0)
#define PWM1_REGULATOR_EN                       (1 << 1)
#define PWM2_REGULATOR_EN                       (1 << 2)
#define PWM3A_REGULATOR_EN                      (1 << 3)
#define PWM3B_REGULATOR_EN                      (1 << 4)
```

Power supply considerations:

- Determine whether to use pwm-regulator based on the external hardware circuit design; it must match the hardware.

2.3 Wakeup Configuration

Configuration items:

```
rockchip,wakeup-config = <...>;
```

Configuration source:

```
// Supports wake-up from all interrupts
#define RKPM_CLUSTER_L_WKUP_EN                  (1 << 0)
// Supports wake-up from all interrupts assigned to the big cores
#define RKPM_CLUSTER_B_WKUPB_EN                 (1 << 1)
// Supports GPIO wakeup
#define RKPM_GPIO_WKUP_EN                       (1 << 2)
// Supports SDIO wakeup
#define RKPM_SDIO_WKUP_EN                       (1 << 3)
// Supports SDMMC wakeup
#define RKPM_SDMMC_WKUP_EN                      (1 << 4)
// Supports TIMER wakeup
#define RKPM_TIMER_WKUP_EN                      (1 << 6)
// Supports USB hot-plug wake-up
#define RKPM_USB_WKUP_EN                        (1 << 7)
// Supports SOFTWARE wakeup
#define RKPM_SFT_WKUP_EN                        (1 << 8)
// Supports WDT wakeup
#define RKPM_WDT_M0_WKUP_EN                     (1 << 9)
// Supports TIMEOUT wakeup, typically used for debugging
#define RKPM_TIME_OUT_WKUP_EN                   (1 << 10)
```

```
// Supports PWM wake-up
#define RKPM_PWM_WKUP_EN                        (1 << 11)
// Supports PCIe wakeup
#define RKPM_PCIE_WKUP_EN                       (1 << 13)
// Supports USB protocol wake-up
#define RKPM_USB_LINESTATE_WKUP_EN              (1 << 14)
```

Wake-up source considerations:

Interrupts not registered with enable_irq_wake() to the GIC during the kernel stage cannot wake up the system.

2.4 Debug Configuration

Configuration options:

```
rockchip,sleep-debug-en = <...>;
```

Debug considerations:

- Setting the value to 1 enables the debug function, which prints ATF suspend and resume logs during the suspend-resume cycle.

2.5 APIO Suspend Configuration

Configuration item:

```
rockchip,apios-suspend = <...>;
```

Configuration source:

```
/* APIO voltage domain */
#define RKPM_APIO0_SUSPEND                      (1 << 0)
#define RKPM_APIO1_SUSPEND                      (1 << 1)
#define RKPM_APIO2_SUSPEND                      (1 << 2)
#define RKPM_APIO3_SUSPEND                      (1 << 3)
#define RKPM_APIO4_SUSPEND                      (1 << 4)
#define RKPM_APIO5_SUSPEND                      (1 << 5)
```

APIO Configuration Notes:

The GPIOs on the RK3399 are grouped into APIO1 through APIO5. When supported by the hardware circuitry, each APIO domain can be powered down independently during suspend.

2.6 GPIO-Controlled Power

Configuration Options:

```
rockchip,power-ctrl = <...>
```

Configuration example:

```
// Pull GPIO1_C1 high during suspend to cut off external power
rockchip,power-ctrl = <&gpio1 RK_PC1 GPIO_ACTIVE_HIGH>;
```

3. Log Output

The following section briefly explains the meaning of Trust log messages during system suspend and resume. For clarity, some log entries are split across multiple lines in the examples below. Different suspend power modes will also produce different log outputs; actual displayed logs shall prevail.

RK3399 System Suspend Logs:

```
// Sleep mode
INFO:    sleep mode config[0xde]:
INFO:            AP_PWROFF
INFO:            SLP_ARMPD
INFO:            SLP_PLLPD
INFO:            DDR_RET
INFO:            SLP_CENTER_PD
// Supported wake-up sources
INFO:    wakeup source config[0x804]:
INFO:            GPIO interrupt can wakeup system
INFO:            PWM interrupt can wakeup system
// PWM regulator that needs to be controlled during suspend
INFO:    PWM CONFIG[0x4]:
INFO:            PWM: PWM2D_REGULATOR_EN
// APIO required for suspend control
INFO:    APIOS info[0x0]:
INFO:            not config
// Power controlled via GPIO
INFO:    GPIO POWER INFO:
INFO:            GPIO1_C1
INFO:            GPIO1_B6
// Value of the sleep mode Register
INFO:    PMU_MODE_CONG: 0x1466bf51
```

RK3399 System Wake-up Log:

```
// Wake-up print
INFO:    RK3399 the wake up information:
INFO:    wake up status: 0x4
INFO:           GPIO interrupt wakeup
INFO:           GPIO0: 0x0
INFO:           GPIO1: 0x200000
INFO:           GPIO2: 0x0
INFO:           GPIO3: 0x0
INFO:           GPIO4: 0x0
// Wakeup source
GPIO interrupt wakeup
gpio1_c5 Interrupt Wakeup System
```